



CS 307: Systems Practicum

Course Code	307
Course Name	Systems Practicum
L-T-P-C	0-0-3-2
Prerequisites	IC150, CS 207
Intended for	B. Tech
Distribution	Compulsory for CSE; CS elective for EE and ME
One-time approval	recommended by the 9th BoA (Oct 2015)

Objective

This course calls for a sequence of 3 Practicum courses for CSE viz. CS 207 Applied Databases Practicum CS 307 Systems Practicum and CS 308 Large Applications Practicum. The erstwhile CS 211 Networks and Database Practicum included both networks and databases tools and programming. Now, the networks material is shifted to CS 307 and CS 207 focuses only on database applications.

Course Outline

Understanding of architecture and working of networks and OS through use of system calls monitoring tools (ps, vmstat, netstat, tcpdump, ping, traceroute..), 'simple socket programming in C. Building multi-process, distributed applications using scripting or other languages. Setting up' and configuring various network and OS services such as web, DNS, email, LDAP, etc. Installing Linux—master boot record, boot loader, disk partitioning, virtualisation (Virtual Box). OS and network security, Techniques for performance measurement. 'data analysis and presentation.

Course Contents

- Architecture and functions of an OS. layered network architecture.
- Processes and threads.
- Concept of name, address, location —IP addresses and DNS; well-known services
- Inter-process communication: files, shared memory, pipes, sockets Synchronisation: lock files, semaphores.
- Packet-switching: packet formats, hop-by-hop routing, routing tables.
- Files, devices file systems.
- Installation and boot procedures of an OS; virtualisation

Lab assignments(9-10 weekly assignments—3 hours in the lab, preceded by at least 3 hours at home— plus a mini-project for 4 Weeks).

Assignment problems are 'designed to expose the students to the following sequence of concepts, skills and tools:

- Create and synchronize processes and threads. Utilities: ps, vmstat, /proc. Bulk data transfer using various forms of IPC. 'Use of networking utilities. ping, traceroute, netstat, tcpdump, /proc
- Socket programming in C: applications such as file transfer, simple chat ping, etc using UDP or TCP. Use of socket options.
- Socket programming in C (continued).
- Program access to i-nodes and directory nodes; Use of sync.
- Virtualisation using eg. VirtualBox; installation of at least 2 different OS'es on virtual machines. Network connections between these OS'es.
- Configuration of DNS NAT NFS, syslogd I/O to serial or parallel port. Use of iotool -Measurement of performance integral to most of the lab exercises: design of experiments measurement techniques Confidence intervals analysis of results, presentation in graphs and tables.
- Mini-projects involving one or more of the following (4 weeks): Email, RMI, PKI, DNS, LDAP/single-sign on, configuring IP routing, _lptables backup/restore of file. systems, Apache, Squid Linux user administration, system logging using syslog, NFS automounting, etc.

References

1. Wikipedia, man pages and online documentation on assorted utilities, tools, components and platforms
2. Ba'utt's, Dawson & Purdy, **Linux Network Administrator's Guide**, 3rd Edition, O'Reilly, 2005.
3. Adelstein & Lubanovic, **Linux System Administration**, O'Reilly, 2007, Kaufmann, Elsevier 2011. (Reference)
4. A.S. Tanenbaum, **Modern Operating Systems**, 3rd Edition, Pearson Ed. Inc., 2008
5. B. Davie & L. Peterson, **Computer networks**, 4th Edition, Morgan and Kaufman, 2011.
6. R. Jain, **The Art of Computer Systems Performance Evaluation**, Wiley, 2008.
7. R.L. Schwartz, T. Phoenix & b.d. foy, **Learning Perl**, 5th Edition O'Reilly, 2008.
8. L. Wall, T. Christiansen & J. Orwant, **Programming Perl**, 3rd Edition, O'Reilly, 2000.